

Notes: Snyder (JPE, 1990)
**Campaign Contributions as Investments: The U.S. House of
Representatives, 1980–1986**

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1 Big picture

- **Research question.** Can special-interest campaign contributions be understood as investments in contingent political favors? More specifically, do economic-interest contributions to U.S. House candidates behave like payments for claims that pay off only if the candidate wins?
- **Core idea.** The paper treats the market for investor contributions and political favors as a simple asset market. A promised favor is a contingent claim: it has value only in the state in which the candidate wins office.
- **Main equilibrium prediction.** If candidates in a race sell the same amount of favors, then a candidate's share of investor contributions should equal his probability of winning.
- **Empirical setting.** The paper studies open-seat U.S. House races in 1980, 1982, 1984, and 1986. Open seats are useful because neither candidate has incumbent seniority, making it more plausible that both can credibly sell similar quantities of favors.
- **Main finding.** The simple asset-market model fits investor PAC contributions surprisingly well, especially after excluding the unusual 1980 election. In the main subsample, the Democratic share of investor contributions is close to an unbiased estimate of the probability that the Democrat wins.
- **Important exception.** Very lopsided races show evidence of demand-side behavior: candidates who are very likely to win appear to accept fewer contributions, consistent with not always selling the maximum number of favors.
- **Contrast with other contributions.** Ideological contributions, individual-plus-candidate contributions, and total contributions do not satisfy the same equilibrium relationship. This supports the paper's distinction between investor-contributors and consumer-contributors.
- **Economic takeaway.** Some campaign contributions are not best interpreted as inputs that mechanically produce votes. They can also be equilibrium prices of political favors, which complicates attempts to estimate the causal effect of campaign spending on electoral success.

2 Incremental contribution of the paper

- **A market-equilibrium view of campaign finance.** Rather than treating contributions only as electoral resources, the paper models special-interest contributions as investments in contingent claims.
- **Clear theoretical restriction.** The model delivers a stark and testable relationship: the share of investor contributions should equal the candidate's probability of winning when both candidates can sell the same amount of favors.
- **Separating contributor motives.** Snyder distinguishes investor-contributors from consumer-contributors. Investor-contributors seek private benefits; consumer-contributors

include ideological groups, individuals, parties, and candidates themselves.

- **Open-seat design choice.** The focus on open-seat House races removes incumbency and seniority differences that would otherwise make one candidate able to sell more favors than the other.
- **Empirical test of equilibrium, not production.** The paper emphasizes that regressions of election outcomes on contributions can describe equilibrium relationships, not structural vote-production functions.
- **Contribution to identification debates.** Snyder gives a theoretical reason why estimating the marginal product of campaign spending is difficult: money and win probabilities are jointly determined in equilibrium.
- **Policy and political-economy insight.** The evidence suggests that economic-interest PACs behave differently from ideological and individual contributors, so aggregate contribution totals can mask very different economic motives.

3 Data and measurement

3.1 Election sample

The empirical analysis uses open-seat races for the U.S. House of Representatives in the elections of 1980, 1982, 1984, and 1986.

- **Initial scope.** The paper considers Democratic and Republican candidates in general-election open-seat contests.
- **Sample restrictions.** Races are dropped when a third-party candidate receives more than 10 percent of the vote or when the winner receives more than 95 percent of the vote.
- **Final sample.** The resulting sample has 161 contests.
- **Why open seats?** Incumbents may be able to sell more favors because of experience, seniority, and committee power. Open seats make the equal-favors assumption more plausible.

3.2 Contribution data

Contribution data come from Federal Election Commission records and cover both the election year and the year before the election.

- **Investor contributions.** Snyder measures investor contributions as the sum of PAC contributions from corporations, labor unions, trade/membership/health organizations, and cooperatives.
- **Ideological contributions.** Nonconnected PAC contributions are used as the measure of ideological contributions.

- **Individual-plus-candidate contributions.** Because early FEC reports did not consistently report all individual giving, Snyder uses total contributions minus PAC and party contributions. In later years this is approximately individual contributions plus candidate contributions.
- **Normalization.** Contribution variables are normalized to account for the growth in real and nominal campaign contributions over time.

3.3 Key variables

- **Democratic share of investor contributions.**

$$x_{iD} = \frac{X_{iD}}{X_{iD} + X_{iR}},$$

where X_{iD} and X_{iR} are investor contributions to the Democratic and Republican candidates in race i .

- **Democratic win indicator.** $W_{iD} = 1$ if the Democratic candidate wins and zero otherwise.
- **Party strength.** A district-level measure based on the Democratic presidential vote share in the district relative to the national Democratic presidential vote share in previous presidential elections.
- **Coattail.** A dummy for presidential-election years in which Republican presidential coattails are expected to help Republican House candidates; it equals one in 1980 and 1984.

3.4 Descriptive patterns

- Investor contributions are only a minority of total campaign contributions, about 22 percent in the summary statistics.
- The average Democratic share of investor contributions is 0.510, while the Democratic candidate wins 47.2 percent of the races.
- The contribution data contain substantial variation across races and across contribution types, which allows the paper to compare investor, ideological, individual, and total contributions.

4 The model

4.1 Investor contributions as contingent claims

Two candidates, D and R , compete for office. Investor-contributors are interested in private favors. Candidate D can promise favors with total monetary value G_D , but these favors are

delivered only if D wins. If p_D is the probability that D wins, equilibrium in the investor-contribution market implies

$$p_D G_D = r X_D,$$

where X_D is total investor contributions to D and r is the required return on comparable investments. Similarly,

$$(1 - p_D) G_R = r X_R.$$

4.2 Equal favors: the central prediction

If both candidates sell the same amount of favors, $G_D = G_R = G$, then combining the two equilibrium conditions gives

$$p_D = \frac{X_D}{X_D + X_R} = x_D.$$

- The probability that a candidate wins equals his share of investor contributions.
- Total investor contributions in a race are constant across races, because $X_D + X_R = G/r$.
- Investor contributions should not be systematically higher in close races if all candidates maximize contributions and sell the same quantity of favors.

4.3 Unequal favors

If one candidate can sell more favors than the other, the candidate selling more favors receives a contribution share that exceeds his probability of winning. In general,

$$p_D = \frac{x_D G_R}{x_D G_R + (1 - x_D) G_D}.$$

Thus, if $G_D > G_R$, then $p_D < x_D$. The extra contribution share compensates investors for the higher supply of favors that D is selling.

4.4 Demand for contributions

The supply-side equilibrium does not by itself specify how many favors candidates choose to sell. Snyder studies two possibilities.

- **Contribution maximization.** Candidates always sell the maximum amount of favors they can credibly deliver.
- **Nonmaximization by likely winners.** Candidates who are very likely to win may not sell all possible favors, perhaps because raising money and delivering favors are costly or unpleasant. Candidates in close races or likely losers still sell all available favors.

4.5 Hypotheses

Hypothesis 1. If all open-seat candidates can sell the same amount of favors and all candidates maximize investor contributions, then:

1. investor contributions to a candidate are an increasing linear function of his probability of winning;
2. the candidate's share of investor contributions equals his probability of winning;
3. total investor contributions are the same across races.

Hypothesis 2. If one type of candidate can sell more favors than the other and all candidates maximize contributions, then the high-favor candidate's investor contribution share exceeds his probability of winning, and total investor contributions are higher when that type is more likely to win.

Hypothesis 3. If likely winners sometimes do not maximize contributions, then Hypothesis 1 holds in relatively close races, but in lopsided races the less likely candidate receives a share of investor contributions greater than his win probability.

5 Empirical specifications

5.1 Testing whether contribution share equals win probability

The key test uses the election outcome equation

$$W_{iD} = \alpha_0 + \alpha_1 x_{iD} + \varepsilon_i.$$

If $p_{iD} = x_{iD}$, then the model predicts

$$\alpha_0 = 0, \quad \alpha_1 = 1.$$

Because W_{iD} is binary, the error variance under the null is $x_{iD}(1 - x_{iD})$. Snyder therefore estimates a corrected version of the regression by dividing observations by

$$v_i = [x_{iD}(1 - x_{iD})]^{1/2}.$$

5.2 Testing for heteroskedasticity

The paper also checks whether the heteroskedasticity correction works. Using residuals from the corrected model, Snyder estimates regressions of squared residuals on x_{iD} and x_{iD}^2 . If the hypothesis is correct, the corrected residuals should not display systematic heteroskedasticity.

5.3 Exclusion tests with political variables

A stronger version of the test adds variables that should predict the Democratic candidate's probability of winning, such as party strength and coattails:

$$W_{iD} = \alpha_0 + \alpha_1 x_{iD} + \beta' Z_i + \varepsilon_i.$$

If investor contribution share already equals the win probability, then Z_i should not add explanatory power and $\beta = 0$.

An equivalent specification regresses the forecast error directly:

$$W_{iD} - x_{iD} = \alpha_0 + \beta' Z_i + \varepsilon_i.$$

5.4 Testing whether total investor contributions are constant

Part iii of Hypothesis 1 predicts that total investor contributions should not vary systematically with the closeness of the race. Snyder tests this using

$$\log(X_{iD} + X_{iR}) = \phi_0 + \phi_1 \log(x_{iD}) + \phi_2 \log(1 - x_{iD}) + \varepsilon_i.$$

Under the constant-total-contribution prediction,

$$\phi_1 = \phi_2 = 0.$$

If ϕ_1 and ϕ_2 are positive, then total investor contributions are higher in close races.

5.5 Comparing contribution types

The same tests are applied to ideological contributions, individual-plus-candidate contributions, and total contributions. The investor-contribution model should fit investor PAC contributions, but it should not necessarily fit contributions motivated by ideology, consumption, party support, or electoral influence.

6 Identification and interpretation

6.1 What the model identifies

The paper does not estimate a structural electoral production function. Instead, it tests equilibrium restrictions implied by an asset-market model of investor contributions. The central claim is that investor contributions and win probabilities are jointly determined in equilibrium.

6.2 Why open-seat races matter

Open-seat races make it more plausible that candidates have similar quantities of favors to sell. In races with incumbents, an incumbent may be able to sell more valuable favors because of seniority, committee assignments, experience, and connections.

6.3 The role of 1980

The 1980 election appears unusual. Republican presidential coattails were stronger than many observers anticipated, and the all-years sample rejects the model once party strength and coattails are added. When 1980 is excluded, the model fits much better.

6.4 Interpreting investor contribution shares

The result $p_D = x_D$ does not mean contributions cause winning with a coefficient of one. It means that in equilibrium investor contributors allocate money as if the candidate's contribution share reflects his probability of winning.

6.5 Why consumer contributions differ

Ideological groups and individuals may contribute for reasons unrelated to buying favors. Ideological contributors may try to help candidates with compatible policy views; individuals and candidates may derive consumption value from campaigning or participation. Therefore, the investor asset-pricing logic should not apply to these contributions.

6.6 Identification warning for spending regressions

Many campaign-finance studies regress votes or victories on spending and interpret the coefficient as the marginal effect of money. Snyder's model warns against this interpretation: if contributions respond to win probabilities, then contribution shares are equilibrium objects, not exogenous campaign inputs.

7 Tables and figures

7.1 Figures 1 and 2: Theoretical relationships between win probability and contribution share

Read Figure 1:

- Figure 1 shows the case in which candidate D sells more favors than candidate R , so $G_D > G_R$.
- The solid curve lies below the 45-degree line: for a given investor contribution share x_D , the win probability p_D is lower.
- Economic interpretation: the high-favor candidate must receive more investor contributions relative to his win probability to make investors indifferent.

Read Figure 2:

- Figure 2 illustrates Hypothesis 3, where likely winners sometimes do not maximize contributions.
- In the middle region $[1 - p_0, p_0]$, contribution share and win probability coincide.
- In lopsided regions, likely winners sell fewer favors, so the less likely candidate receives a contribution share greater than his win probability.

Economic message: The slope and curvature of the relationship between p_D and x_D reveal whether candidates sell equal favors, whether one type sells more favors, or whether likely winners do not fully maximize investor contributions.

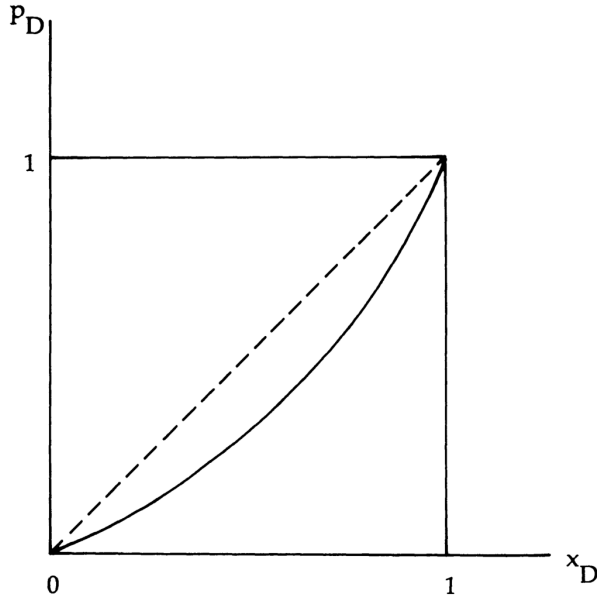


FIG. 1.— $G_D > G_R$

Figure 1: Unequal favors, $G_D > G_R$.

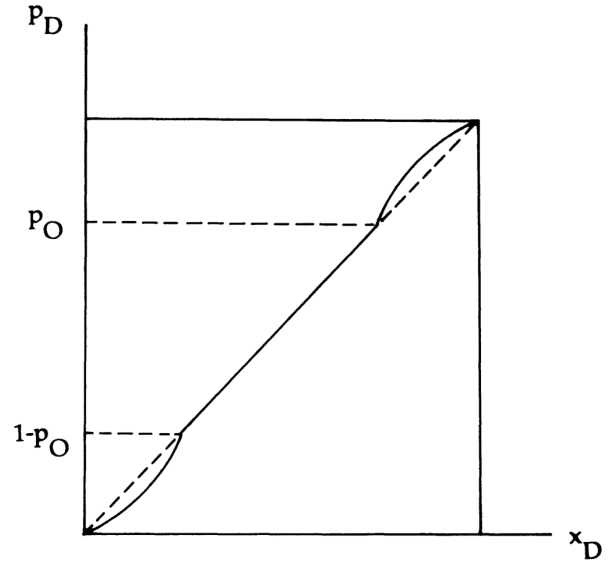


FIG. 2

Figure 2: Likely winners do not always maximize contributions.

7.2 Table 1: Summary statistics

Read:

- Table 1 reports normalized contribution totals, investor PAC contributions, ideological contributions, individual-plus-candidate contributions, the Democratic win indicator, and party strength.
- The sample contains 161 open-seat races.
- Average Democratic total contributions are about 425,877, while average Republican total contributions are about 471,905.
- Average Democratic investor contributions are 95,931, and average Republican investor contributions are 102,928.

- The Democratic share of investor contributions averages 0.510, while Democrats win 47.2 percent of races.

Economic message: Investor contributions are a meaningful but minority component of campaign finance. The near equality between the Democratic investor share and Democratic win frequency motivates the paper’s central equilibrium test.

TABLE 1
SUMMARY STATISTICS, 1980–86 ($n = 161$)

	Mean	Standard Deviation
DEM TOTAL CONTRIBUTIONS	425,877	328,428
REP TOTAL CONTRIBUTIONS	471,905	299,081
DEM INVESTOR CONTRIBS ($X_{i,D}$)	95,931	61,768
REP INVESTOR CONTRIBS ($X_{i,R}$)	102,928	75,461
DEM SHARE INVEST CONTRIBS ($x_{i,D}$)	.510	.295
DEM IDEOLOGICAL CONTRIBS	18,590	20,258
REP IDEOLOGICAL CONTRIBS	24,187	18,858
DEM INDIVIDUAL + CANDIDATE CONTRIBS	302,955	297,721
REP INDIVIDUAL + CANDIDATE CONTRIBS	319,057	232,488
DEM WIN*	.472	.501
PARTY STRENGTH†	-.013	.090

NOTE.—To control for the increase in real and nominal contributions, contributions (of all types) in year t are multiplied by (average DEM + REP TOTAL CONTRIBUTIONS in 1986)/(average DEM + REP TOTAL CONTRIBUTIONS in year t).

* Equals one if the Democrat won, and equals zero if the Republican won.

Table 1: Summary statistics, 1980–1986.

7.3 Table 2: Investor-contribution model, full sample

Read:

- Table 2 estimates the basic model using investor contributions for 1980–1986.
- The corrected linear specification has an intercept close to zero and a slope close to one.
- The joint hypothesis $\alpha_0 = 0$ and $\alpha_1 = 1$ is not rejected in the corrected model.
- Piecewise-linear tests also do not reject the simple relationship $p_D = x_D$.
- The residual tests indicate that the heteroskedasticity correction largely works for investor contributions.

Economic message: The most basic evidence supports the investor-contribution asset-market prediction: investor contribution shares behave like win probabilities.

		LINEAR		PIECEWISE LINEAR (3)
		Uncorrected (1)	Corrected (2)	
A. Dependent Variable: DEM WIN				
Intercept	α_0	-.122* (.064)	-.027 (.039)	.010 (.046)
DEM SHARE INVEST CONTRIBS	α_1	1.162*** (.112)	.989*** (.054)	...
DEM SHARE INVEST CONTRIBS IN (0, .5)	β_1	...	...	.373 (.320)
DEM SHARE INVEST CONTRIBS IN (.5, 1)	β_2	...	...	1.92*** (.425)
DEM SHARE INVEST CONTRIBS IN (0, 1)	β_3	...	...	.507 (.311)

Adjusted R^2	.41	.85	.85
Mean squared error	.15	1.12	1.10
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \beta_1 = \beta_2 = 1$
F-statistic	1.885	1.062	1.819
Prob > F	.155	.348	.128
B. Dependent Variable: SQUARED RESIDUALS from Regressions Above			
Intercept	γ_0	-.022 (.044)	1.253 (1.243) (1.233)
DEM SHARE INVEST	γ_1	.896*** (.187)	-4.192 (5.248) (5.203)
CONTRIBS			
DEM SHARE INVEST	γ_2	-.858*** (.174)	5.923 (4.869) (4.828)
CONTRIBS) ²			
Adjusted R^2		.13	.01
Hypothesis test	$H_0: \gamma_1 = \gamma_2 = 0$	$H_0: \gamma_1 = \gamma_2 = 0$	$H_0: \gamma_1 = \gamma_2 = 0$
F-statistic	12.239	1.696	1.698
Prob > F	.001	.187	.187

* Significantly different from zero at the .10 level.
*** Significantly different from zero at the .01 level.

* Significantly different from zero at the .10 level.
*** Significantly different from zero at the .01 level.

Table 2, part 1: Basic linear and piecewise tests.

Table 2, part 2: Residual and hypothesis tests.

7.4 Tables 3 and 4: Controls and the 1980 outlier

Read Table 3:

- Table 3 adds party strength, coattails, and year indicators to the investor-contribution model.
- In the full sample, party strength and coattails add explanatory power, so the strict model is rejected.
- The negative coattail result is largely driven by 1980, when Republican coattails were stronger than many actors anticipated.

Read Table 4:

- Table 4 estimates the corrected linear model separately by election year.
- The model is rejected for 1980 but not for 1982, 1984, or 1986.
- In 1980, Democratic candidates appear to have received too many investor contributions relative to their true probabilities of winning.

Economic message: Investor contribution markets appear to price win probabilities well in normal years, but unexpected macro-political shocks can make contribution shares inaccurate ex post.

		DEPENDENT VARIABLE			
		DEM WIN		DEM WIN - DEM SHARE INVEST CONTRIBS	
		(1)	(2)	(3)	(4)
Intercept	α_0	.107***	.042	.001	-.045
DEM SHARE INVEST	α_1	.090	.050	.050	.040
CONTRIBS		.810***	.814***	...	...
PARTY STRENGTH	β_1	.072	.070	...	...
COATTAIL	β_2	.772***	.920***	.340*	.402***
YEAR 1980	β_3	.140	.230	1.181	1.102
YEAR 1982	β_4	...	...	...	...
YEAR 1984	β_5	...	...	...	...
YEAR 1986	β_6	...	...	...	...
Adjusted R^2		.86	.86	.86	.86
Mean squared error		1.05	.87	.87	.87
Hypothesis test		4.54	4.80	5.79	4.47
F-statistic		.002	.000	.012	.001
Prob > F					

* Significantly different from zero at the .10 level.
** Significantly different from zero at the .05 level.
*** Significantly different from zero at the .01 level.

Table 3: Investor-contribution model with political controls.

TABLE 4
ESTIMATES OF THE (CORRECTED) LINEAR MODEL, BY YEAR
Dependent Variable: DEM WIN

		1980 (n = 35)	1982 (n = 54)	1984 (n = 24)	1986 (n = 39)
Intercept	α_0	-.081	.008	-.004	-.062
DEM SHARE	α_1	.173	.059	.070	.053
INVEST CONTRIBS		.846***	1.034***	.960**	1.069***
Adjusted R^2		.218	.077	.145	.075
Mean squared error		.66	.92	.74	.93
Hypothesis test		1.98	.73	1.17	.65
F-statistic		3.847	.565	.077	.682
Prob > F		.032	.572	.926	.512

** Significantly different from zero at the .05 level
*** Significantly different from zero at the .01 level

Table 4: Corrected linear model by election year.

7.5 Table 5: Investor-contribution model excluding 1980

Read:

- Table 5 repeats the Table 2 tests for 1982–1986, excluding 1980.
- The corrected linear model again has an intercept near zero and a slope near one.
- The piecewise-linear model does not reject equality between investor contribution share and win probability.
- Residual tests are consistent with the maintained heteroskedasticity correction.

Economic message: Once the unusually surprising 1980 election is removed, the basic investor-contribution model fits especially well.

TABLE 5
MODELS FOR INVESTOR CONTRIBUTIONS, 1982–86 (n = 117)

		LINEAR		PIECEWISE LINEAR
		Uncorrected (1)	Corrected (2)	(3)
A. Dependent Variable: DEM WIN				
Intercept	α_0	-.091 (.075)	-.020 (.033)	.007 (.039)
DEM SHARE INVEST CONTRIBS	α_1	1.130** (.128)	1.041** (.049)	...
DEM SHARE INVEST CONTRIBS IN (0, 50)	β_1	...	...	.589** (.305)
DEM SHARE INVEST CONTRIBS IN (50, 100)	β_2	...	...	1.504*** (.415)
DEM SHARE INVEST CONTRIBS IN (0, 1)	β_3	...	...	.954*** (.295)

Table 5, part 1: Models for investor contributions, 1982–1986.

Adjusted R^2	.40	.91	.91
Mean squared error	.15	.78	.78
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \beta_1 = \beta_2 =$
F-statistic	.752	.352	.739
Prob > F	.474	.704	.568
B. Dependent Variable: SQUARED RESIDUALS from Regressions Above			
Intercept	γ_0	-.010 (.436)	.660 (.430)
DEM SHARE INVEST CONTRIBS	γ_1	1.000*** (.185)	2.000 (1.742)
DEM SHARE INVEST CONTRIBS ²	γ_2	-1.033*** (.175)	-2.7002 (1.641)
Adjusted R^2		.23	.03
Hypothesis test	$H_0: \gamma_1 = \gamma_2 = 0$	$H_0: \gamma_1 = \gamma_2 = 0$	$H_0: \gamma_1 = \gamma_2 = 0$
F-statistic	18.066	2.566	2.241
Prob > F	.0001	.081	.111

** Significantly different from zero at the .05 level.
*** Significantly different from zero at the .01 level.

Table 5, part 2: Residual tests for the 1982–1986 sample.

7.6 Tables 6 and 7: Controls and total investor contributions, excluding 1980

Read Table 6:

- Table 6 shows that after excluding 1980, party strength and coattails do not significantly improve the investor-contribution model.
- When investor contributions are omitted, party strength and coattails become good predictors of Democratic victory.
- This is exactly what the model predicts if investor contribution share already embeds the relevant win-probability information.

Read Table 7:

- Table 7 tests whether total investor contributions are constant across races.
- Total investor contributions are higher in closer races, rejecting the strict constant-total prediction of Hypothesis 1.
- The result suggests that demand effects matter in at least some races, even though the share prediction works well.

Economic message: The contribution-share prediction is strong, but total investor contributions reveal that closeness affects the level of economic-interest giving.

TABLE 6
MODELS FOR INVESTOR CONTRIBUTIONS, 1982–86 ($n = 117$)

		DEPENDENT VARIABLE		
		DEM WIN (1)	DEM WIN – DEM SHARE INVEST CONTRIBS (2)	DEM WIN (3)
Intercept	α_0	.019 (.047)	.002 (.023)	.572*** (.045)
DEM SHARE INVEST CONTRIBS	α_1	.971*** (.070)	...	...
PARTY STRENGTH	β_1	.331 (.239)	.264 (.173)	2.701*** (.451)
COATTAIL	β_2	-.018 (.054)	-.012 (.051)	-.202** (.099)
Adjusted R^2		.91	.77	.18
Mean squared error		91	86	22.58
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1, \beta_1 = \beta_2 = 0$			
F-statistic		.664	.477	.0001
Prob > F		.618		

** Significantly different from zero at the .05 level.

*** Significantly different from zero at the .01 level.

Table 6: Investor-contribution model with controls, 1982–1986.

TABLE 7
MODELS FOR TOTAL INVESTOR CONTRIBUTIONS, 1982–86
Dependent Variable: TOTAL (DEM + REP) INVEST CONTRIBS (Log)

		OBSERVATIONS WITH DEM SHARE INVEST CONTRIBS IN THE INTERVAL		
		(0, 1) ($n = 117$) (1)	[1, 9] ($n = 90$) (2)	[15, 85] ($n = 82$) (3)
Intercept	δ_0	12.722*** (.095)	13.340*** (.257)	13.002*** (.405)
DEM SHARE INVEST CONTRIBS (log)	δ_1	.263*** (.049)	.695*** (.197)	.487* (.272)
REP SHARE INVEST CONTRIBS (log)	δ_2	.339*** (.046)	.666*** (.165)	.417 (.257)
Adjusted R^2		.32	.15	.01
Hypothesis test	$H_0: \delta_1 = \delta_2 = 0$			
F-statistic		27.837	9.137	1.612
Prob > F		.0001	.0002	.205

* Significantly different from zero at the .10 level.

*** Significantly different from zero at the .01 level.

Table 7: Total investor contributions and race closeness.

7.7 Tables 8 and 9: Ideological and individual-plus-candidate contributions

Read Table 8:

- Table 8 applies the same tests to ideological contributions.
- The model $p_D = y_D$, where y_D is the Democratic ideological-contribution share, is rejected.
- Party strength matters, and residual behavior differs from what the investor model predicts.

Read Table 9:

- Table 9 applies the tests to individual-plus-candidate contributions.
- The investor model is again rejected.
- Likely losers receive too many individual-plus-candidate contributions relative to their probabilities of winning, consistent with consumption motives in running for office or giving to candidates.

Economic message: The model is not a generic model of all campaign contributions. It is specifically a model of economic-interest contributions that resemble investments in favors.

MODELS FOR IDEOLOGICAL CONTRIBUTIONS, 1982-86 (n = 109)				
DEPENDENT VARIABLE				
	DEM WIN (1)	DEM WIN (2)	DEM WIN - DEM SHARE IDEOL CONTRIB (3)	
Intercept	α_0	.078*	.279***	.113***
		(.045)	(.059)	(.038)
DEM SHARE IDEOL CONTRIBS	α_1	.949***	.612***	...
		(.094)	(.109)	
PARTY STRENGTH	β_1	...	2.336***	1.3330***
			(.455)	(.380)
COATTAIL	β_2	...	...	.026
			(.082)	(.085)
Adjusted R^2		.67	.73	.13
Mean squared error		1.43	1.17	1.50
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \alpha_1 = 1, \beta_1 = \beta_2 = 0$	$H_0: \alpha_0 = 0, \beta_1 = \beta_2 = 0$	
F-statistic		1.956	7.732	5.442
Prob > F		.146	.000	.002
Dependent Variable: SQUARED RESIDUALS from Regressions Above				
Intercept	γ_0	2.552***	...	...
		(.438)		
DEM SHARE IDEOL CONTRIBS	γ_1	-2.740***	...	...
		(.884)		
Adjusted R^2		.07		

* Significantly different from zero at the .10 level.

*** Significantly different from zero at the .01 level.

Table 8: Models for ideological contributions.

DEPENDENT VARIABLE				
	DEM WIN	DEM WIN	DEM WIN - DEM SHARE IND + CAND CONTRIBS	
Intercept	α_0	-.062	.076***	.097***
		(.047)	(.079)	(.033)
DEM SHARE IND + CAND CONTRIBS	α_1	1.133***	1.047***	...
		(.096)	(.138)	
PARTY STRENGTH	β_1	...	.945**	1.054***
			(.436)	(.294)
COATTAIL	β_2	...	-.211***	-.203***
			(.751)	(.071)
Adjusted R^2		.74	.77	.13
Mean squared error		.74	.66	.66
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \alpha_1 = 1, \beta_1 = \beta_2 = 0$	$H_0: \alpha_0 = 0, \beta_1 = \beta_2 = 0$	
F-statistic		2.094	5.214	6.967
Prob > F		.128	.001	.000
Dependent Variable: SQUARED RESIDUALS from Regressions Above				
Intercept	γ_0	.163	...	...
		(.320)		
DEM SHARE IND + CAND CONTRIBS	γ_1	3.613**	...	...
		(1.467)		
(DEM SHARE IND + CAND CONTRIBS) ²	γ_2	-4.169***	...	...
		(1.525)		
Adjusted R^2		.05		
Hypothesis test	$H_0: \gamma_1 = \gamma_2 = 0$			
F-statistic		4.924		
Prob > F		.001		

** Significantly different from zero at the .05 level.

*** Significantly different from zero at the .01 level.

Table 9: Models for individual-plus-candidate contributions.

7.8 Table 10: Total contributions

Read:

- Table 10 applies the same tests to total contributions.
- The model that contribution share equals win probability is clearly rejected for total contributions.
- Total contributions combine investor motives, ideological motives, party support, individual consumption motives, and candidate self-financing.

Economic message: Aggregating all contributions hides the equilibrium pricing logic of investor contributions. The asset-market model is most informative when applied to the subset of contributions with clear economic-interest motives.

DEPENDENT VARIABLE				
	DEM WIN (1)	DEM WIN (2)	DEM WIN - DEM SHARE TOTAL CONTRIBS (3)	
Intercept	α_0	-.950***	.014	.078**
		(.046)	(.071)	(.032)
DEM SHARE TOTAL CONTRIBS	α_1	1.256***	1.137***	...
		(.093)	(.136)	
PARTY STRENGTH	β_1	...	.655	.978***
			(.426)	(.283)
COATTAIL	β_2	...	-.196***	-.180***
			(.072)	(.070)
Adjusted R^2		.76	.78	.12
Mean squared error		.65	.60	.60
Hypothesis test	$H_0: \alpha_0 = 0, \alpha_1 = 1$	$H_0: \alpha_0 = 0, \alpha_1 = 1, \beta_1 = \beta_2 = 0$	$H_0: \alpha_0 = 0, \beta_1 = \beta_2 = 0$	
F-statistic		3.801	4.924	6.231
Prob > F		.025	.001	.001
Dependent Variable: SQUARED RESIDUALS from Regressions Above				
Intercept	γ_0	-.206	...	...
		(.207)		
DEM SHARE TOTAL CONTRIBS	γ_1	4.718***	...	...
		(.923)		
(DEM SHARE TOTAL CONTRIBS) ²	γ_2	-5.097***	...	...
		(.952)		
Adjusted R^2		.19		
Hypothesis test	$H_0: \gamma_1 = \gamma_2 = 0$			
F-statistic		14.333		
Prob > F		.000		

** Significantly different from zero at the .05 level.

Table 10: Models for total contributions.

8 Short summary

- The paper models special-interest campaign contributions as investments in contingent political favors.
- A favor promised by a candidate pays off only if the candidate wins office.
- No-arbitrage logic implies that, if both candidates sell the same amount of favors, a candidate's share of investor contributions should equal his probability of winning.
- The empirical analysis uses open-seat U.S. House races from 1980, 1982, 1984, and 1986.
- Investor contributions are measured using economically oriented PAC categories: corporations, labor unions, trade/membership/health organizations, and cooperatives.
- The basic investor-contribution model fits the data surprisingly well, especially after excluding 1980.
- The 1980 election is an outlier because Republican presidential coattails were stronger than expected, causing investor contribution shares to misprice Democratic candidates' true win probabilities.
- Total investor contributions are higher in close races, so the strict prediction that total investor contributions are constant is rejected.
- Ideological, individual-plus-candidate, and total contributions do not satisfy the same model.
- The main empirical lesson is that investor PAC contributions behave differently from other forms of campaign finance.
- The main methodological lesson is that regressions of electoral success on spending may estimate equilibrium relationships rather than causal production functions.

9 Conclusion

9.1 Core conclusion

Snyder shows that economic-interest campaign contributions in U.S. House open-seat races can be interpreted as investments in political favors. Despite the simplicity of the model, the aggregate behavior of investor PAC contributions is broadly consistent with the asset-market prediction that a candidate's share of investor contributions equals his probability of winning.

9.2 Economic intuition

Investor-contributors care about expected private benefits. A favor promised by a candidate is valuable only if the candidate wins, so investors discount the favor by the candidate's

probability of victory. In equilibrium, contributions adjust so that expected returns on favors are equalized across candidates.

9.3 Incremental contribution revisited

The paper’s key contribution is to shift the interpretation of special-interest contributions from campaign inputs to equilibrium prices. This reframes campaign finance as a market in contingent claims and yields sharp empirical restrictions.

9.4 Implications

The findings imply that not all campaign contributions should be pooled together. Economic-interest PAC contributions can reflect investment motives, while ideological and individual contributions can reflect very different motives. For empirical work, this means that total spending or total contributions are hard to interpret without a model of who gives and why.

9.5 Limitations

The setting is limited to open-seat House races from 1980–1986. Incumbent races may differ because incumbents can sell more valuable favors. The model is intentionally stark and does not fully explain total contribution levels, especially the fact that investor contributions are higher in close races. The evidence is also based on equilibrium restrictions rather than random variation in contributions.

9.6 Takeaway

The enduring takeaway is that campaign contributions can be equilibrium prices of political access and favors, not simply exogenous campaign resources. For investor PACs, contribution shares appear to reveal market beliefs about win probabilities; for other contributors, different motives dominate.